FORMATIVE ASSESSMENT – 3 (FA-3)

MATHEMATICS • CLASS 9 • SLIP TEST

Question 1

Find four different solutions for the linear equation **2x + y = 7**.

Question 2

If the point **(3, 4)** lies on the graph of the equation **3y = ax + 7**, find the value of **a**.

Question 3

In an isosceles triangle **ABC**, **AB = AC**. If the altitude **AD** is drawn perpendicular to side **BC**, prove that **AD** bisects **BC** (that is, **BD = CD**).

Question 4

Two line segments **AB** and **CD** bisect each other at point **O**. Prove that **ΔAOC ≅ ΔBOD** and show that **AC** is parallel to **BD**.

Question 5

The angles of a triangle are in the ratio **2 : 3 : 4**. Find the measure of each angle of the triangle and classify the triangle based on its angles.

SOLUTIONS & SCHEME OF WORK

CLASS 9 MATHEMATICS • FA-3 SLIP TEST

Solution 1

Rewrite as **y = 7 − 2x**:

• If x = 0, y = 7 − 2(0) = 7 ⇒ (0, 7)

• If x = 1, y = 7 − 2(1) = 5 ⇒ (1, 5)

• If x = 2, y = 7 − 2(2) = 3 ⇒ (2, 3)

• If x = 3, y = 7 − 2(3) = 1 ⇒ (3, 1)

Ans: Four solutions are (0, 7), (1, 5), (2, 3), and (3, 1)

Solution 2

Substitute x = 3 and y = 4 into **3y = ax + 7**:

3(4) = a(3) + 7

12 = 3a + 7 ⇒ 3a = 12 − 7 ⇒ 3a = 5 ⇒ **a = 5 / 3**

Ans: a = 5 / 3

Solution 3

In right-angled triangles **ΔADB** and **ΔADC**:

• ∠ADB = ∠ADC = 90° (Given: AD ⊥ BC)

• AB = AC (Given: Hypotenuses are equal)

• AD = AD (Common side)

By RHS congruence criterion: **ΔADB ≅ ΔADC**.

Therefore, BD = CD by CPCTC. Hence, AD bisects BC.

Ans: Proved BD = CD (AD bisects BC)

Solution 4

Since segments bisect each other at O: **OA = OB** and **OC = OD**.

In ΔAOC and ΔBOD: OA = OB, ∠AOC = ∠BOD (vertically opposite angles), OC = OD.

By SAS congruence rule: **ΔAOC ≅ ΔBOD**.

Thus, ∠OAC = ∠OBD (CPCTC). Since these alternate interior angles are equal, AC ∥ BD.

Ans: ΔAOC ≅ ΔBOD verified and AC ∥ BD proved

Solution 5

Let the three angles be **2x, 3x, and 4x**. The angle sum of a triangle is 180°:

2x + 3x + 4x = 180° ⇒ 9x = 180° ⇒ **x = 20°**

• First angle: 2(20°) = **40°**

• Second angle: 3(20°) = **60°**

• Third angle: 4(20°) = **80°**

Since all angles are less than 90°, it is an **Acute-angled triangle**.

Ans: Angles are 40°, 60°, 80° (Acute-angled triangle)